

White box Testing:

⇒ Also called glass box, clear-box, open-box or structural testing.

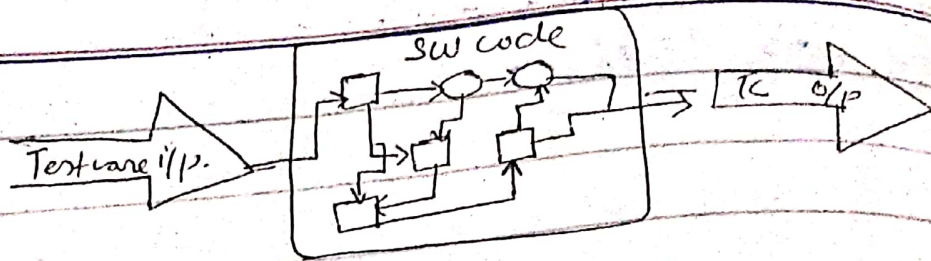
⇒ The white box testing technique is concerned with testing the internal structure of a software & its implementation.

⇒ The software tester uses this testing technique to examine:

- The internal structure of the code of system software.
- The integration of modules among them and with external system or subsystems.
- The flow of inputs and outputs through the software system.

⇒ WB Testing can be applied at the unit, integration & system levels of the software process testing.

⇒ Purpose: whether the software produces the expected results or outcomes.



⇒ Using the white-box testing technique the following major tests are performed on the software system.

Control flow testing:

⇒ To examine logical paths within a module at the unit level, paths between modules at the integration level and between subsystems at the system level.

Dataflow testing:

⇒ Performed to examine all data variables included in the code.

⇒ it is tested where the variables are declared & they are used.

⇒ Using white-box testing, a software tester can derive test cases to ensure that:

- All independent paths within a module are exercised at least once.

- All logical decisions for true or false paths
- All loops executed at their boundaries & within their operational bounds.
- All internal data structures are exercised to ensure their validity.

White box testing technique

1) Code coverage Analysis

- Statement coverage
- Branch coverage
- Path coverage

2) Flow graph & notations

3) Basis path Testing

- Graph matrices
- Independent path
- Cyclomatic Complexity
- Deriving test cases

4) Control Structuring Testing

- DFT
- Condition Testing

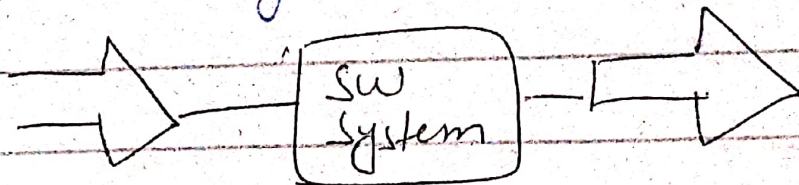
○ Loop Testing

- ↳ Simple
- ↳ nested
- ↳ unisolated

↳ Unstructured loops

Black-box Testing:

- ⇒ Also called functional, behavioral or closed box testing.
- ⇒ Focus on the functional requirements of the software system.
- ⇒ Tester examines the functionality or behavior of the software system without knowing its internal structure or coding.
- ⇒ Internal structure or implementation details are not visible to the tester.
- ⇒ So, in the eyes of the tester, system is treated as black-box that's why it is called BBT.
- ⇒ can be applied during later stages such as system, acceptance or integration testing.
- ⇒ Functionality or behavior of the software system can be tested by giving I/P & examining the returned outputs.



+ List of errors

Black-box Testing Techniques:

- 1) Graph based methods
 - directed links
 - undirected links
 - Bidirectional links
 - Parallel links
- 2) Equivalence Partitioning
- 3) Boundary value Analysis